

Pulmotor pt – 60

****Summary of Experiment: Study and Sketch of Reviving Apparatus (Model: Pulmotor PT-60 and Carevent)****

****Introduction****

The Pulmotor PT-60 was originally developed by Heinrich Dräger in 1906 and has since undergone significant modifications. It is designed for automatic cycling of respiration, delivering an oxygen-air mixture to a patient's lungs while also expelling CO₂-laden air. This apparatus can function as both a resuscitator and an inhalator.

****Constructional Features****

The Pulmotor PT-60 comprises two steel oxygen cylinders with a capacity of 2.5 liters each, pressurized to 200 kg/cm². Key components include:

- A pressure-reducing valve that decreases high-pressure oxygen to 5 kg/cm².
- A distribution manifold with three shut-off valves and a safety valve.
- A cycling valve box (central to its operation).
- An oxygen economizer breathing bag, an inhalation facepiece, and three oral airways.
- Additional elements like a secretion bottle, aspiratory injector, and an inhalation mask.

Each of the three shut-off valves serves specific functions: one connects to the aspirating injector, another to the inhalation mask, and the third to the cycling valve box. The cycling valve box regulates the automatic cycling of breathing phases.

****Working and Operating Process****

The Pulmotor operates in cycles, shifting between inspiration and expiration phases:

1. ****Inspiration Phase****: Oxygen flows through an injector nozzle, drawing in air through a regulating valve. The oxygen-air mixture is then delivered to the patient. As the lungs fill, positive pressure builds up in the valve box, activating the control membrane and initiating the switch to the expiration phase.

2. ****Expiration Phase****: The injector sucks air from the patient's lungs, expelling it outside. When lung pressure reaches a negative threshold, the control membrane is triggered, and the cycle returns to the inspiration phase.

A change knob allows manual adjustment to initiate the expiration phase.

****Procedure and Safety Notes****

1. Ensure the apparatus is properly observed to maintain a consistent respiratory rhythm.
2. During resuscitation, if heart activity stops, perform heart massage during the expiration phase.
3. If CO poisoning is suspected, loosen the tongue as needed.
4. Avoid using the Pulmotor, except as an inhalator, in cases of nitrous fume poisoning.

****Key Points****

Immediate manual resuscitation is critical if spontaneous breathing ceases, as the first few minutes are vital.

CareVent

Introduction

The O-Two CAREvent® CA is a sophisticated, automated resuscitator designed to be operated with minimal input in hazardous environments, where rapid response is crucial for patient survival. The CAREvent® CA requires only a single control for operation, streamlining the resuscitation process in critical scenarios by reducing the rescuer's workload.

Constructional Features and Specifications

The CAREvent® CA operates without batteries or an electrical supply, relying on micro-pneumatic circuitry. Key specifications include:

- Tidal Volume: 500 ml per breath
- Breathing Rate: 10 breaths per minute, with an inspiratory-expiratory (I

) ratio of 1:2

- Automatic and Manual Flow Rate: 15 L/min
- Positive Pressure Bleed Flow: 2 L/min
- Demand Breathing Flow Rate: Up to 120 L/min, allowing spontaneous breathing by the patient if they regain respiratory function.
- Maximum Airway Pressure: 60 cmH₂O
- Cylinder Duration: Approximately 82 minutes at 10 breaths per minute and a tidal volume of 500 ml

Additional accessories include a head harness for hands-free operation and a PEEP-equipped transport ventilator circuit for use during transport.

Working and Operating Process

The CAREvent® CA operates automatically, providing a pre-set volume and breathing rate suitable for resuscitation:

1. **Automatic Ventilation:** When oxygen/air is turned on, the device initiates ventilation at 10 breaths per minute with a tidal volume of 500 ml.
2. **Demand Breathing Feature:** If the patient begins to breathe independently, the automatic cycling stops, allowing the patient to control their own respiratory rate and volume. If they stop breathing again, automatic cycling resumes without intervention from the rescuer.
3. **Manual Override:** A manual override button allows the rescuer to deliver two breaths and perform 30 chest compressions, following current CPR guidelines. This button can also supplement spontaneous breaths if needed.
4. **Mask Purge:** The device provides a continuous 2 L/min bleed flow through the mask, reducing the risk of ambient toxic air being inhaled if the mask seal leaks.

Procedure and Safety Notes

1. Ensure the oxygen supply is turned on and the mask is properly secured to the patient's face.
2. Observe for spontaneous breathing, which will halt automatic ventilation; if the patient stops breathing, the device will automatically resume cycling.
3. Use the manual override during CPR or as needed for additional ventilations.
4. For transport, accessories like the gurney clip can secure the device, while the PEEP port maintains appropriate airway pressure.

Key Points

The O-Two CAREvent® CA's simplicity of operation and automatic cycling are essential for emergency use in hazardous environments, making it a reliable option for quick and effective resuscitation without requiring extensive rescuer intervention.

BG – 174A

Introduction

The Dräger BG-174A is a closed-circuit breathing apparatus designed for use in high-risk environments, providing a regulated supply of oxygen (O₂) in a self-contained system that minimizes external air exposure. Manufactured by Dräger AGCO, Germany, it is used primarily in scenarios requiring sustained oxygen supply under challenging conditions, with a safe operational period of up to 4 hours for strenuous activity.

Main Specifications

- **O₂ Cylinder Capacity:** 2 liters (empty); 400 liters at 200 kg/cm².
- **Breathing Bag Capacity:** 6 liters.
- **Weight (fully charged):** 12.8 kg.
- **O₂ Flow Rate:** Constant 1.5 liters/min, with additional flow as governed by the user's lung demand.
- **CO₂ Absorbent:** 1.5 kg of anhydrous NaOH.

Working and Operating Process

The BG-174A uses a closed-circuit system that recycles exhaled air by removing CO₂ and enriching it with fresh oxygen, allowing for a sustainable breathing environment. The process operates as follows:

1. **Exhalation:** Air exhaled by the user passes through the exhalation tube and valve into a CO₂ absorption canister, where CO₂ is removed using anhydrous NaOH.
2. **Purification and Recycling:** The purified air flows into the breathing bag, where it is stored and ready for reuse.
3. **Inhalation:** During inhalation, the air is drawn from the breathing bag, enriched with O₂ by the lung-demand valve, and delivered to the user.
4. **Pressure Regulation:** Excess pressure in the system is controlled by an automatic relief valve, preventing buildup. The apparatus also has a pre-flushing device that automatically clears nitrogen (N₂) from the breathing circuit with a 7-liter burst of O₂ when the cylinder valve is opened.

Special Features

- **Lung-Demand Valve:** This diaphragm-operated valve responds to the user's inhalation needs, delivering additional oxygen as required.
- **Pre-Flushing Device:** This feature initiates a burst of O₂ when the cylinder is first opened, ensuring a pure oxygen flow from the start.
- **Cooling Option:** A dry ice cooler can be used instead of the saliva trap to lower inhaled air temperature to approximately 15°C.

Procedure and Safety Notes

1. **Start-Up:** Open the cylinder valve to activate the system, which will pre-flush the breathing circuit to eliminate nitrogen.
2. **Monitoring:** Observe the pressure gauge and shut-off valve to ensure a consistent oxygen supply.
3. **Safety Checks:** Verify CO₂ absorption efficiency with regular checks and, if necessary, replace the anhydrous NaOH canister.

4. **Temperature Control:** For hot environments, the optional dry ice cooler helps maintain comfortable inhaled air temperatures.

Key Points

The Dräger BG-174A's closed-circuit design, coupled with precise oxygen flow control and CO₂ absorption, enables long-duration respiratory support in high-risk environments.

BG – 4

Introduction

The Dräger BG-4 is a closed-circuit, compressed oxygen (O₂) breathing apparatus used for prolonged respiratory protection, commonly in hazardous environments. Certified by NIOSH in 1995, it comes in both 3-hour and 4-hour versions and provides positive pressure to prevent external contaminants from entering the system. It utilizes a soda-lime canister to absorb exhaled CO₂.

Specifications

- **O₂ Flow Rate:** Constant minimum flow of 1-5 liters per minute, with additional volume delivered as needed via a demand valve.
- **Operating Duration:** 3-hour (fiberglass cylinder version) and 4-hour (standard O₂ cylinder) configurations.
- **CO₂ Absorption:** Disposable soda-lime canister.
- **Face Mask:** Same as used in the BG-174A.
- **Cooling Mechanism:** A heat-absorbent canister with ice to cool inhaled air.

Working and Operating Process

The BG-4 apparatus operates in a closed circuit, continuously recycling the user's exhaled air by removing CO₂ and providing fresh oxygen. Its main operating stages are as follows:

1. **Exhalation:** The user exhales into the face mask, passing exhaled air through the exhalation check valve, breathing hose, CO₂ absorbent canister, and into the spring-loaded breathing bag. The soda-lime canister absorbs CO₂ from exhaled air.
2. **CO₂ Absorption and Purification:** The relief valve purges excess exhaled air to prevent pressure buildup, ensuring that purified air flows efficiently.
3. **Inhalation:** On inhalation, purified air is drawn from the breathing bag, routed through the heat-absorbent canister (which contains ice for cooling) and inhalation hose, and back into the face mask.
4. **Cooling System:** Before use, a large ice cylinder is placed in the heat-absorbent canister. This cools the inhaled air, improving user comfort during prolonged usage.

Procedure and Safety Notes

1. **Preparation:** Insert the ice cylinder in the heat-absorbent canister to enable inhaled air cooling. Ensure the soda-lime canister is installed correctly.
2. **Startup:** Open the O₂ supply to initiate a constant flow, allowing the apparatus to reach optimal positive pressure.
3. **Monitoring:** Regularly check pressure levels and ensure the relief valve operates correctly to maintain safe breathing conditions.
4. **Post-Use:** Dispose of the soda-lime canister, as it is single-use only, and replace it for future operations.

Key Points

The Dräger BG-4 is a robust closed-circuit apparatus ideal for extended operations in toxic environments. Its constant oxygen supply, CO₂ absorption system, and cooling mechanism enhance safety and comfort, supporting up to 4 hours of usage in its positive-pressure configuration.

Self – rescuers

1. Filter-Type Self-Rescuer (MSA IW65)

Description and Purpose

The MSA IW65 Self-Rescuer is a portable, single-use, air-purifying respirator for emergency protection against carbon monoxide (CO) in underground mines, especially after explosions or fires. The device does not supply oxygen but instead converts CO into non-toxic CO₂, allowing for safe passage through contaminated areas. It is effective only if the surrounding oxygen levels are at least 16%, as it cannot function in severely oxygen-deficient atmospheres.

Specifications

- Weight: Approx. 1 kg
- Duration: 60 minutes
- Principle of Operation: The device contains a Hopcalite filter (manganese dioxide and copper oxide) that catalyzes the conversion of CO to CO₂, generating heat in the process. Air is cooled via a heat exchanger before inhalation, with additional filters for dust and moisture protection.

Procedure

Upon detecting fire or explosion, the user should:

1. Activate the respirator immediately.
2. Place the respirator mouthpiece in the mouth and secure the nose clip.
3. Breathe through the device, which filters CO and provides cooled air.

Limitations and Drawbacks

The device does not supply oxygen and can be uncomfortable due to the heat generated during CO

filtration. It is only effective in environments with adequate oxygen (above 16%) and will not protect in the presence of other toxic gases.

2. Chemical Oxygen Self-Rescuer (RakshaKavach)

Description and Purpose

The RakshaKavach is a closed-circuit, chemical oxygen self-rescuer designed to produce oxygen chemically for emergency use in mines, submarines, and other industries. It operates independently of ambient air, making it suitable for escape in both oxygen-deficient and toxic environments. The unit provides 30 minutes of respiratory protection and is for single-use only.

Components

- **KO₂ Canister:** Generates oxygen through a chemical reaction with CO₂ and water vapor.
- **Breathing Bag:** A 6-liter bag with a relief valve to store purified air.
- **Mouthpiece and Nose Clip:** Ensures breathing through the device.
- **Heat Protection:** Prevents discomfort from reaction-generated heat.

Working Principle

Activated by the user's breathing, the KO₂ in the canister reacts with exhaled CO₂ and water vapor to produce oxygen. This reaction is demand-sensitive, with more oxygen generated as breathing intensifies. The reaction by-products, such as potassium carbonate (K₂CO₃), retain CO₂, ensuring the wearer breathes CO₂-free oxygen.

Procedure

1. Remove the unit from its belt.
2. Open and prepare the device following step-by-step instructions.
3. Attach the mouthpiece and secure the nose clip.
4. Breathe normally to fill the breathing bag with fresh oxygen.
5. Adjust the straps and wear the helmet before moving to a safe location.

Precautions

- Only trained personnel should use RakshaKavach.
 - The unit should only be opened in emergencies, as tampering may reduce effectiveness.
 - Do not remove the mouthpiece until reaching a safe environment.
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These devices provide critical respiratory protection, converting harmful gases or generating oxygen for safe evacuation in emergencies. The self-rescuer is suitable for CO-rich environments with sufficient oxygen, while RakshaKavach provides oxygen and is effective in oxygen-deficient and toxic atmospheres.

Both require proper training and strict adherence to operational protocols to ensure effectiveness and safety during use.

RAM – 1

The RAM-1 is a portable, real-time aerosol dust monitor that measures airborne respirable dust concentrations in environments such as mines. Its sensing principle is based on near-forward scattered electromagnetic radiation in the near-infrared (940 nm), emitted by a pulsed gas semiconductor diode. The scattered radiation is detected by a silicon diode with a low-noise preamplifier, and signal processing uses a synchronous lock-in scheme for stable, drift-free readings.

Specifications

- **Dust Concentration Ranges:** Selectable from 0-2, 20, or 200 mg/m³ for precise measurements over five decades.
- **Time Constants:** Four selectable response times (0.5, 2.8, and 32 s).
- **Flow Rate:** Total sampling flow rate of 2 liters/minute; a secondary purge flow of 0.2 liters/minute keeps optical surfaces clean.
- **Display and Alerts:** A digital LCD shows real-time dust concentration, with indicators for calibration (flashing "K") and low battery voltage ("VDC").
- **Battery Life:** 6 to 8 hours on a full charge, with indefinite operation when connected to an AC power supply.

Procedure

1. **Power On:** Switch on the instrument using the top switch.
2. **Range and Time Setting:** Select the desired concentration range and time constant for measurements.
3. **Position Inlet Valve:** Move the inlet valve to the sample position (downward).
4. **Purging:** Begin in the "CLEAR" position for 1 minute to purge air.
5. **Flow Checks:** Observe total and purge flow rates on the flowmeter (2.2 liters/minute total, 0.22 liters/minute for purge).
6. **Adjust Flow:** Use the flow screw to adjust the total flow rate, and the valve to adjust the purge flow rate.
7. **Sampling:** Place the instrument in sampling mode, and observe dust concentration readings on the LCD.
8. **Over-Range Indication:** If dust exceeds the selected range, "1" with blanked zeros will appear on the display.

9. **Protection During Use:** Close the protective panel to shield the display and controls.

Application

The RAM-1 provides direct, real-time measurements of respirable dust in environments with the following concentration ranges:

- **0-2 mg/m³**
- **2-20 mg/m³**
- **20-200 mg/m³**

This experiment demonstrates how to operate the RAM-1 to monitor dust concentrations accurately, ensuring reliable data collection in occupational settings. The instrument's real-time functionality and selectable range settings make it suitable for continuous, on-site dust monitoring and environmental safety assessments.

Gravimetric Dust Sampler

The MRE 113A Gravimetric Dust Sampler is a self-powered, portable device used for measuring respirable airborne dust in mining environments by gravimetric sampling, a method that relies on the difference in weight before and after dust collection. The instrument is mounted at a height of two-thirds of the mine roadway and operates for a full shift to collect dust particles on a filter. By comparing the initial and final weights of the filter, the airborne dust concentration (in mg/m³) is calculated.

Specifications and Working

- **Flow Rate:** 2.5 liters per minute (l/min), with three lines on the flowmeter to indicate correct flow rate (middle line), underflow (bottom line), and overflow (top line).
- **Battery:** Rechargeable 7.5V battery (90mA), powering a diaphragm-type pump to draw dust-laden air.
- **Filter Type:** 5.5 cm diameter glass fiber or membrane filters, with the membrane being more convenient.
- **Size Selector:** The multi-channel horizontal elutriator collects fine particles, while heavier particles settle by gravity.

The instrument consists of a size selector, filter holder, and a diaphragm-type pump. Air is drawn through the size selector, where coarse particles are separated, and fine respirable particles are collected on the filter. The counter registers the amount of air drawn, and an adjustable crank regulates the flow rate, which should remain at 2.5 l/min to ensure proper sampling.

Procedure

1. **Preparation:** A clean filter is weighed with a sensitivity of 0.1 mg and mounted in the filter holder, then secured in the sampler. The fully charged battery is inserted, and the counter is set to zero.

2. **Placement:** The sampler is placed at breathing height in the desired area, usually covering an entire work shift.
3. **Operation:** The sampler runs at 2.5 l/min; the duration can vary based on dust concentration.
4. **After Sampling:** The volume of air sampled is noted. The filter is then removed and reweighed to determine the dust weight.
5. **Calculation:** Airborne respirable dust concentration (mg/m^3) is calculated by dividing the change in filter weight by the volume of air sampled.

Application

This experiment demonstrates the use of the MRE 113A for accurate, quantitative dust concentration measurements in mines. The gravimetric method offers reliable data on harmful airborne dust levels, aiding in compliance with occupational health and safety standards.

Konimeter

The Konimeter is a dust sampling instrument designed for quick spot readings of airborne dust concentration, measured in particles per cubic centimeter (PPCC). It consists of a spring-operated plunger pump, dust deposition chamber, a glass plate, air inlet jet, and a built-in microscope. By taking a rapid air sample and depositing particles onto a prepared glass plate, the Konimeter provides an immediate dust count, useful in various industrial and mining environments.

Specifications and Working

- **Pump:** A spring-operated pump draws a 5 cc air sample by a gun-metal plunger, which moves in a cylinder and locks automatically at the bottom of its stroke.
- **Dust Deposition Chamber:** Contains a rotating glass plate on which dust is deposited, with 30 engraved sectors for multiple samples.
- **Glass Plate:** Circular, with heat-resistant properties, placed at a fixed distance of 0.50 mm from the inlet jet to ensure dust particles land uniformly.
- **Air Inlet Jet:** Carburetor-type jet (0.5 mm diameter) channels the air sample into the chamber.
- **Microscope:** Magnifies up to $\times 216$ for viewing dust particles, with a graticule fitted in the eyepiece for rapid evaluation.

The instrument functions by drawing air through the jet, depositing particles on a treated glass plate with petroleum jelly to capture dust. The microscope allows easy dust counting using intersecting lines and sectors on the glass plate.

Procedure

1. **Preparation:** Clean the glass plate and apply a thin layer of petroleum jelly on the non-engraved side. Place the plate back in position.

2. Operation: Depress the plunger until it locks, then position the Konimeter and lift the air inlet cover. Turn the ring to release the plate for upstroke and begin sample collection.
3. Dust Count Calculation: After sampling, the dust count on the plate is recorded. Dust concentration is calculated by scaling the number of particles seen in 5 cc to a 1 cc sample using a formula.

Limitations and Accuracy

- Results may vary between different Konimeter models and under varying conditions.
- High dust concentrations can complicate readings.
- Accuracy is affected by the nature of the adhesive film on the plate.
- High impact velocity may disrupt dust aggregates, affecting measurement accuracy.
- Only suitable for spot sampling, not continuous monitoring.

Application

This experiment provides a simple method to measure dust concentration in air, especially suitable for quick checks in industrial environments. However, the Konimeter's use is limited by accuracy issues and conditions that may alter sample results.